

Diagrammatic equation for the operator $\hat{S}[0,1]$:

The left side shows a vertical line with two horizontal lines crossing it, each marked with an 'X' at the intersection. This is equal to a square box containing two arcs connecting the top and bottom horizontal lines. This is equal to the operator $\hat{S}[0,1]$, which is defined as a sum over j and k from 0 to $d-1$ of the tensor product of states $|j\rangle\langle k|$ and $|k\rangle\langle j|$.

$$= \text{Diagram} \iff \hat{S}[0,1] = \sum_{j,k=0}^{d-1} |j\rangle\langle k| \otimes |k\rangle\langle j|$$